# linearAlgebra: Internal Tests

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## Tests

```
#> test-linearAlgebra-chol-dot
#> Test passed with 3 successes.

#> test-linearAlgebra-d-mat-dot
#> Test passed with 3 successes.

#> test-linearAlgebra-d-mat
#> Test passed with 2 successes.

#> test-linearAlgebra-d-of-mat-dot
#> Test passed with 1 success.

#> test-linearAlgebra-d-of-vec-dot
#> Test passed with 2 successes.

#> test-linearAlgebra-deviation
#> Test passed with 4 successes.

#> test-linearAlgebra-diag-of-vech-dot
#> Test passed with 4 successes.
```

```
#> test-linearAlgebra-diff-cols-dot

#> Test passed with 3 successes.

#> test-linearAlgebra-ldl

#> Test passed with 4 successes.

#> test-linearAlgebra-p-inv-dmat

#> Test passed with 2 successes.

#> test-linearAlgebra-pinv-of-d-mat-dot

#> Test passed with 4 successes.

#> test-linearAlgebra-positive-definite-2-test-dot

#> Test passed with 3 successes.

#> test-linearAlgebra-positive-definite-test-dot

#> Test passed with 3 successes.

#> test-linearAlgebra-positive-semi-definite-test-dot

#> Test passed with 6 successes.
#> Test passed with 1 success.

#> test-linearAlgebra-symmetric-test-dot

#> Test passed with 2 successes.

#> test-linearAlgebra-symof-vech-2-dot

#> Test passed with 2 successes.

#> test-linearAlgebra-symof-vech-dot

#> Test passed with 9 successes.
```

```
#> test-linearAlgebra-symof-vech

#> Test passed with 1 success.

#> test-linearAlgebra-symof-vechs-dot

#> Test passed with 2 successes.

#> test-linearAlgebra-symof-vechs

#> Test passed with 1 success.

#> test-linearAlgebra-test-positive-definite

#> Test passed with 3 successes.

#> test-linearAlgebra-test-positive-semi-definite

#> Test passed with 6 successes.
#> Test passed with 4 successes.
#> Test passed with 1 success.

#> test-linearAlgebra-test-symmetric

#> Test passed with 2 successes.

#> test-linearAlgebra-vec-dot

#> Test passed with 2 successes.

#> test-linearAlgebra-vec

#> Test passed with 1 success.

#> test-linearAlgebra-vech-check

#> Test passed with 4 successes.

#> test-linearAlgebra-vech-dot
```

```
#> Test passed with 2 successes.

#> test-linearAlgebra-vech-names-dot

#> Test passed with 2 successes.

#> test-linearAlgebra-vech

#> Test passed with 1 success.

#> test-linearAlgebra-vechs-check

#> Test passed with 5 successes.

#> test-linearAlgebra-vechs-dot

#> Test passed with 2 successes.

#> test-linearAlgebra-vechs-names-dot

#> Test passed with 1 success.

#> test-linearAlgebra-vechs

#> Test passed with 1 success.
#> [[1]]
#> [[1]][[1]]
#> [[1]][[1]]$value
#> function (program)
#> {
#>     testthat::skip_on_cran()
#>     testthat::skip_on_ci()
#>     testthat::skip_if(condition = Sys.which(names = program) ==
#>         "", message = paste0(program, " is not installed"))
#> }
```

```
#>
#> [[1]][[1]]$visible
#> [1] FALSE
#>
#>
#> [[1]][[2]]
#> [[1]][[2]]$value
#> [[1]][[2]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[2]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[3]]
#> [[1]][[3]]$value
#> [[1]][[3]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[3]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[4]]
#> [[1]][[4]]$value
#> [[1]][[4]]$value[[1]]
```

```
#> [1] TRUE
#>
#>
#> [[1]][[4]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[5]]
#> [[1]][[5]]$value
#> [[1]][[5]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[5]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[6]]
#> [[1]][[6]]$value
#> [[1]][[6]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[6]]$visible
#> [1] TRUE
#>
#>
```

```
#> [[1]][[7]]
#> [[1]][[7]]$value
#> [[1]][[7]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[7]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[8]]
#> [[1]][[8]]$value
#> [[1]][[8]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[8]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[9]]
#> [[1]][[9]]$value
#> [[1]][[9]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[9]]$visible
```

```
#> [1] TRUE
#>
#>
#> [[1]][[10]]
#> [[1]][[10]]$value
#> [[1]][[10]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[10]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[11]]
#> [[1]][[11]]$value
#> [[1]][[11]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[11]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[12]]
#> [[1]][[12]]$value
#> [[1]][[12]]$value[[1]]
#> [1] TRUE
```

```
#>
#>
#> [[1]][[12]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[13]]
#> [[1]][[13]]$value
#> [[1]][[13]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[13]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[14]]
#> [[1]][[14]]$value
#> [[1]][[14]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[14]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[15]]
#> [[1]][[15]]$value
```

```
#> [[1]][[15]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[15]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[16]]
#> [[1]][[16]]$value
#> [[1]][[16]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[16]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[17]]
#> [[1]][[17]]$value
#> [[1]][[17]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[17]]$visible
#> [1] TRUE
#>
```

```
#>
#> [[1]][[18]]
#> [[1]][[18]]$value
#> [[1]][[18]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[18]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[19]]
#> [[1]][[19]]$value
#> [[1]][[19]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[19]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[20]]
#> [[1]][[20]]$value
#> [[1]][[20]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[20]]$visible
```

```
#> [1] TRUE
#>
#>
#> [[1]][[21]]
#> [[1]][[21]]$value
#> [[1]][[21]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[21]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[22]]
#> [[1]][[22]]$value
#> [[1]][[22]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[22]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[23]]
#> [[1]][[23]]$value
#> [[1]][[23]]$value[[1]]
#> [1] TRUE
```

```
#>
#>
#> [[1]][[23]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[24]]
#> [[1]][[24]]$value
#> [[1]][[24]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[24]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[25]]
#> [[1]][[25]]$value
#> [[1]][[25]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[25]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[26]]
#> [[1]][[26]]$value
```

```
#> [[1]][[26]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[26]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[27]]
#> [[1]][[27]]$value
#> [[1]][[27]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[27]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[28]]
#> [[1]][[28]]$value
#> [[1]][[28]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[28]]$visible
#> [1] TRUE
#>
```

```
#>
#> [[1]][[29]]
#> [[1]][[29]]$value
#> [[1]][[29]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[29]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[30]]
#> [[1]][[30]]$value
#> [[1]][[30]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[30]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[31]]
#> [[1]][[31]]$value
#> [[1]][[31]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[31]]$visible
```

```
#> [1] TRUE
#>
#>
#> [[1]][[32]]
#> [[1]][[32]]$value
#> [[1]][[32]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[32]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[33]]
#> [[1]][[33]]$value
#> [[1]][[33]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[33]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[34]]
#> [[1]][[34]]$value
#> [[1]][[34]]$value[[1]]
#> [1] TRUE
```

```
#>
#>
#> [[1]][[34]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[35]]
#> [[1]][[35]]$value
#> [[1]][[35]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[35]]$visible
#> [1] TRUE
```

## Environment

```
ls()


#> [1] "root"
```

## Class

```
#> [[1]]
#> [1] "root_criterion"
```

## References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/